Viscosity of Newtonian Fluids

Final Lab Report
Unit Operations Lab 1
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Group 1
Joshua Olusoji
Omar Zaruk
Sanad Abu Hweij
Qussay Suleiman

1. Executive Summary (10 pts)

This experiment identified the viscosity behavior of a 1 wt% Carbopol solution in water by using a capillary tube viscometer. The purpose was to determine how the solution behaved because it is important to get accurate viscosity data to predict fluid transport in chemical engineering processes.

The approach involved measuring flow through six capillary tubes of different lengths and diameters. For each run a fixed mass and elapsed time was recorded. Flow rates were calculated from volume and time, and plots of shear stress versus shear rate were used to extract the flow behavior. The flow behavior index n ranged between 0.2 and 0.5, with most values around 0.35. Since $n<1$, this shows that the Carbopol solution is shear thinning. Looking at the results, we can say that the Carbopol solution is a non-Newtonian fluid that becomes less viscous as the shear rate increases. This means the material will flow easier when it gets pumped at higher speeds. The shear thinning property is important in industrial processes, because it reduces energy requirements for equipment while maintaining high viscosity at rest.

While the experiment clearly Identified shear-thinning behavior there were limitations because of the equipment precision. The gauge made it difficult to observe pressure drops directly, and assumptions such as constant density and no wall slip added some uncertainty. Even with those errors the flow behavior was consistent across runs which made our conclusion stronger.

Overall, the experiment showed how capillary viscometry can be used to classify something as either Newtonian or Non-Newtonian fluids; in our case, it was a non-Newtonian fluid.

2. Introduction (10 pts)

The objective of this experiment was to characterize the viscosity of a 1 wt% Carbopol solution in water using a capillary tube viscometer. The characterization of viscosity is important to us chemical engineers, as it allows us to predict transport behaviors relating to heat and mass transfer. The experiment apparatus consists of a pressure regulator, pressure gauge, pressure relief valve, viscometer lid, viscometer valve, and capillary tubes. With the capillary tube viscometer, we will be able to measure our fluid's flow rate and pressure drop across various tubes differing in length and diameter. Capillary tube viscometers are used widely in chemical, pharmaceutical, petroleum, and food industries to measure viscosity because of its high accuracy and simple operation.

3. Theory (10 pts)

Describe the theory and principles related to the experiment and its objectives. Briefly describe how the experimental data will be used to obtain your results. Include important equations or empirical correlations. Use references as appropriate.

Viscosity is a key property of fluid mechanics; it's the ratio of the shearing stress on a fluid to the shearing rate. It's the measure of the resistance of a fluid to a shearing force, and can be interpreted as the measure of the momentum flux.

This concept can be easily understood by considering the flow between 2 parallel plates. A velocity profile will be established if one of the plates is moved at a constant velocity. The force required to keep this plate in motion is directly proportional to the viscosity and the velocity gradient. In this this example the only velocity we have is in the x direction as a function of y, $v_x(y)$. Which corresponds to:

$$\tau_{yx} = -\mu \frac{dv_x}{dy} (1)$$

In this case τ_{yx} is the shear stress: the force per unit area of the contacting plate in the +x direction on a surface with a unit normal in the + y direction. The negative sign comes from the convention which align with figure 1 because the velocity gradient is negative

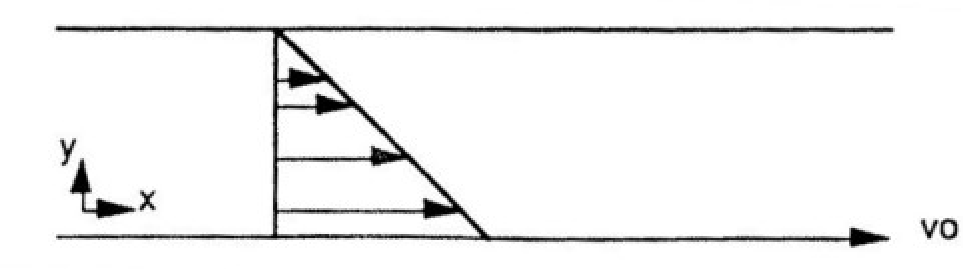


Figure 1: velocity profile between 2 parallel plates

However, experimentally, it's been identified that fluids can be categorized differently. The fluids that follow equation 1 can be categorized as Newtonian fluids, and the ones that don't exhibit this linear relationship between the shear stress and velocity gradient are called Non-Newtonian fluids. For these fluids the viscosity is not constant and is a function of the rate of shear. Figure 2 illustrates these different behaviors of fluids relating the shear stress and the velocity gradient.

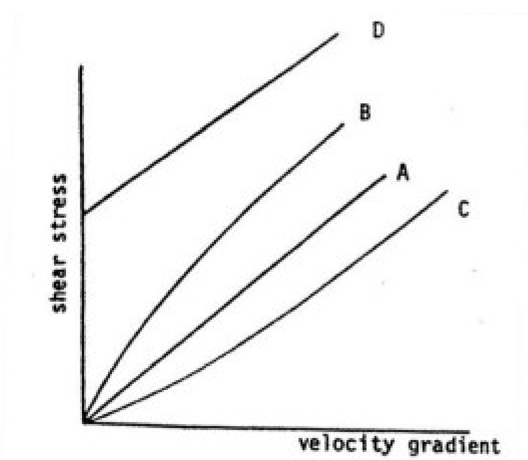


Figure 2. Relationship between strain and shear for different fluids: (A) Newtonian fluid;

(B) psuedo-plastic fluid; (C) dilatant fluid; (D) Bingham fluid.

Curves B and C are for fluids that are called pseudo-plastic and dilatant, respectively. For these fluids a power-law relationship known as the Ostwald-deWaele Model can characterize the of shear rate on the velocity gradient:

$$\tau_{yx} = \pm m \left| \frac{dv_x}{dy} \right|^n (2)$$

Where m and are n are empirical constants and the ±is there so the shear stress is in the correct direction relative to the velocity gradient. Equation 2 can also be rewritten as:

$$\tau_{yx} = -m \left| \frac{dv_x}{dy} \right|^{n-1} \frac{dv_x}{dy} = -\mu_{app} \frac{dv_x}{dy} (3)$$

Where μ_{app} is the apparent viscosity, which is not constant and is a function of the rate of the shear experienced by the fluid.

The actual relationship between the velocity gradient and the shear is correlated with the constants m and n from equation (2).

For n > 1 (shear-thickening), the fluid behavior resembles **curve B** of Figure 2. For n < 1(shear-thinning), the fluid behavior resembles **curve C**. When n = 1, the fluid is Newtonian, and the behavior corresponds to **curve A,** which shows the linear relationship between the shear stress and the velocity gradient.

From the capillary tube viscometer , the flow rate and pressure drop across a capillary tube can be used to measure the viscosity of a fluid.

For a Newtonian fluid, the viscosity is related to the flowrate and pressure drop by the Hagen-Poiseuille equation:

$$Q = \frac{\Delta P \pi R^4}{8 \mu L} (4)$$

Q is the volumetric flow rate, $\Delta P$is the pressure drop across the capillary, R and L are the radius and length, respectively, and $\mu$is the fluid viscosity. The graph of Q vs ΔPwould give a linear line with a slope of $\pi R^4 / 8 \mu L$ From that, the viscosity can be obtained.

For our experiment, we are working with a non-Newtonian fluid, an analytical equation of the flow rate can be obtained from the Ostwald-deWaele viscosity model:

$$m \left[\frac{3n+1}{n} \right]^n * \left(\frac{Q}{\pi R^3} \right)^n = \left(\frac{\Delta P}{2L} R \right) (5)$$

A plot of $\log \left(\Delta \frac{P * R}{2L} \right) vs. \log \left(\frac{Q}{\pi R^3} \right)$ will give a slope of n and an intercept of

$$\log\left\{m\left[\frac{3n+1}{n}\right]^{n}\right\}$$

From the slope and the intercept value, the parameters of m and n can be defined and calculated as a function of the diameter and length of the capillary.

The shear stress at the wall can be calculated using the following formula, which relates the stress to the pressure drop, radius, and length of the capillary tube:

$$\tau_{rz}=\frac{\Delta P}{2L}r$$

The velocity gradient can be obtained using the power law model:

$$\frac{dv_z}{dr}=\left(\frac{1}{m}\frac{\Delta P}{2L}\right)^{1/n}r^{\frac{1}{n}}$$

Where m and n are the empirical constants found experimentally.
[1**]**

With these relationships and the above equations, we can characterize the viscosity of our 1wt% Carbopol solution.

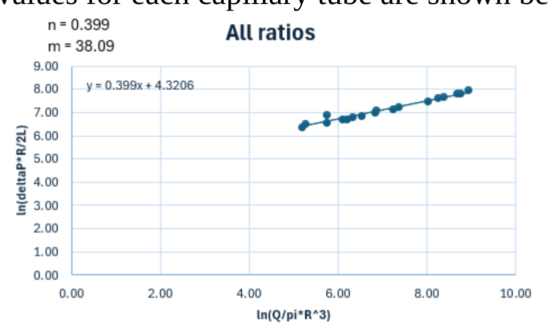
4. Results (15 pts)

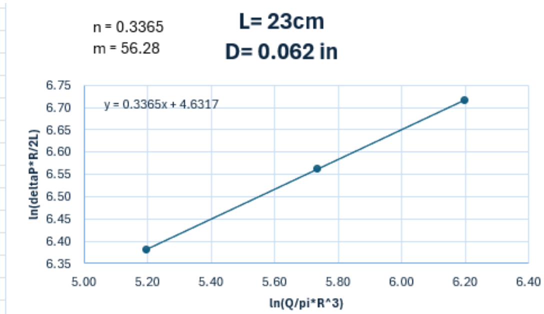
In lab, we characterized flow through six capillary tubes by first measuring each tube's length L and inner diameter D, then setting the desired pressure drop ΔP with a regulator. For every run, we collected a fixed 20 g of Carbopol effluent and recorded the elapsed time and mass (using mass instead of volume to avoid error from the airy gel). Bulk density was measured once via a specific-gravity method. We assumed bulk density was constant across all runs, although in practice pressure, shear, and temperature effects may cause small variations, introducing uncertainty into calculated flow rates. Volume for each run was computed as $V=m/\rho$. This constant density assumption may result in slight error, since all measurements were compiled and converted to SI units prior to analysis. Wall slip and gravity effects were assumed negligible given the relatively short tubes and low elevation.

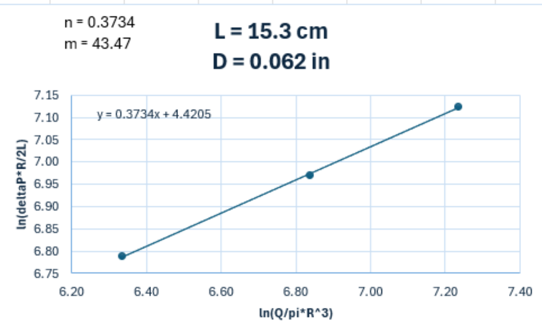
Since it was unknown whether this fluid was Newtonian, we did not use the Hagen–Poiseuille equation to calculate flow; instead, we computed the flow rate as $Q = V/t$, by dividing the Carbopol volume (from mass/density) by the elapsed time for each run. With Q determined, we plotted log(ΔP·R/2L) vs. log(Q/πR³) and used the fitted slope n to identify whether the solution is shear-thinning or shear-thickening.

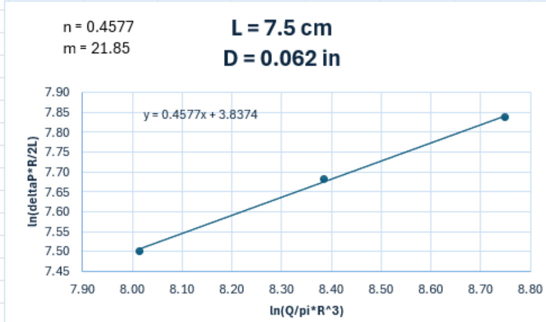
The flow behavior index *n* was obtained from the slope of the log–log plots, while the consistency index *m* came from rearranging the intercept equation. Across the seven graphs created (six for each L/D ratio and one with all data combined), the n-values ranged between 0.2 and 0.5, with most values close to 0.35. This consistently indicates that the Carbopol solution is **shear-thinning**, since $n<1$. In contrast, the calculated m-values varied significantly, falling between about 20 and 200. Such variation is expected because *m* is highly sensitive to measurement conditions, geometry, and data fitting, whereas *n* is generally more robust.

The parameters have clear physical meaning: *n* is dimensionless and represents the degree of shear-thinning or shear-thickening, while *m* has units of $Pa \cdot s^n$ and reflects the fluid's consistency. Different capillaries gave different results primarily because geometry directly sets the flow rate under the same ΔP: a longer tube increases viscous resistance and reduces Q, while a larger diameter provides more cross-sectional area and increases Q. Since all capillaries were tested at the same pressure drops, these length/diameter effects dominate the observed differences in fitted *m*, even when the underlying material behavior is the same. The graphs showing the extracted *n* and *m* values for each capillary tube are shown below, and the L/D ratios can be seen in table 2.









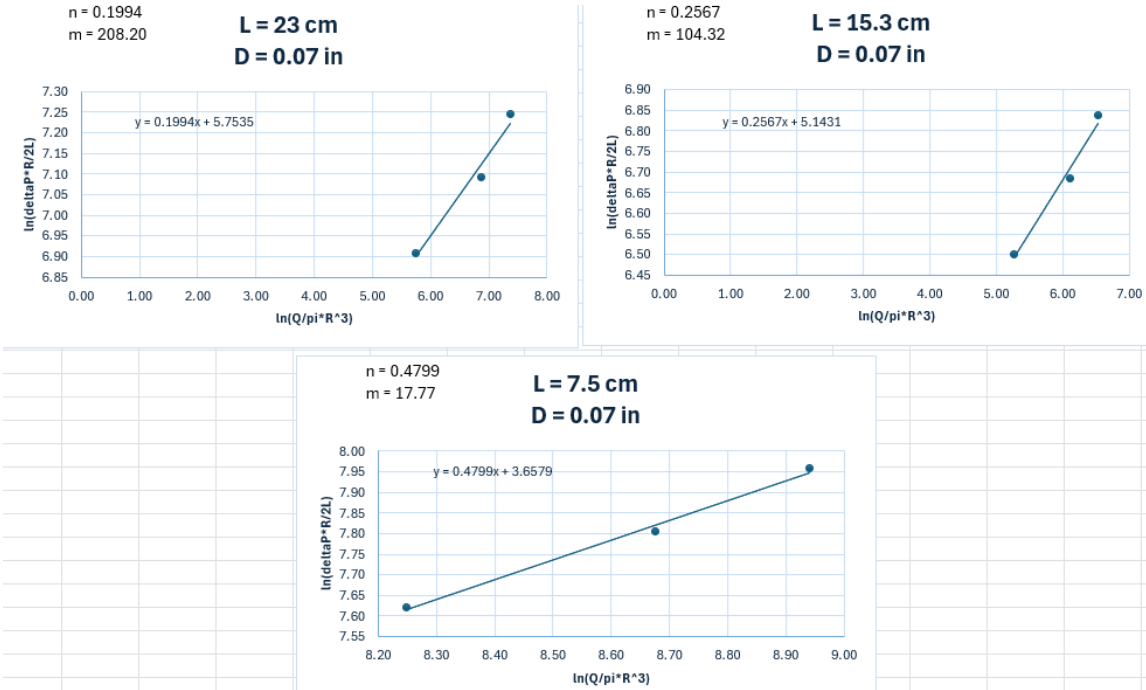


Figure 3: fluid type determination

Shear stress at the wall was calculated using the pressure drop across the tube, radius, and length according to $\tau_W = \frac{\Delta P}{2L} r$. To obtain the velocity gradient, we applied the power-law model, which relates shear stress to the velocity profile as $\tau_{rz} = m \left| \frac{d v_z}{dr} \right|^n$. Rearranging this gives the expression $\frac{d v_z}{dr} = \left(\frac{1}{m} \frac{\Delta P}{2L} \right)^{1/n} r^{1/n}$, allowing the local velocity gradient to be calculated as a function of radial position and the experimentally determined parameters *m* and *n*.

The results were plotted for the two capillary diameters (0.062 in and 0.07 in), with each line representing a different tube length. The data show a clear upward trend of shear stress with velocity gradient, consistent with non-Newtonian behavior. For both diameters, shorter capillaries yield higher shear rates at a given stress, reflecting the reduced resistance to flow. Additionally, the larger diameter (0.07 in) reaches higher shear stresses and shear rates overall than the 0.062-in tube, as the wider passage reduces viscous resistance and allows greater flow under the same pressure drop. The shear stress versus velocity gradient plot shows a decreasing slope at higher shear rates, indicating shear-thinning behavior; this corresponds to a flow behavior index of $n<1$, confirming that the Carbopol solution is a pseudoplastic fluid. The results can be seen on the graphs below.

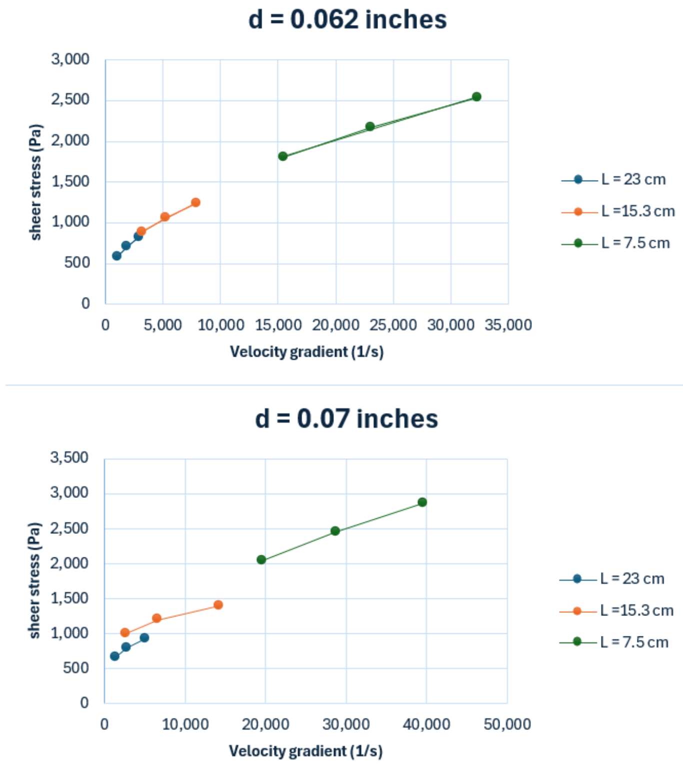
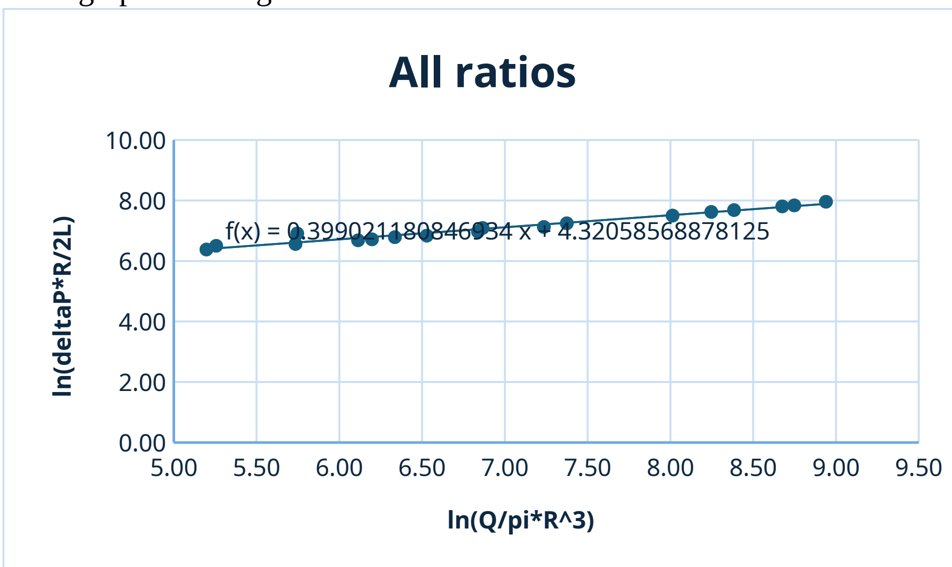


Figure 4: Sheer Stress vs Velocity Gradient

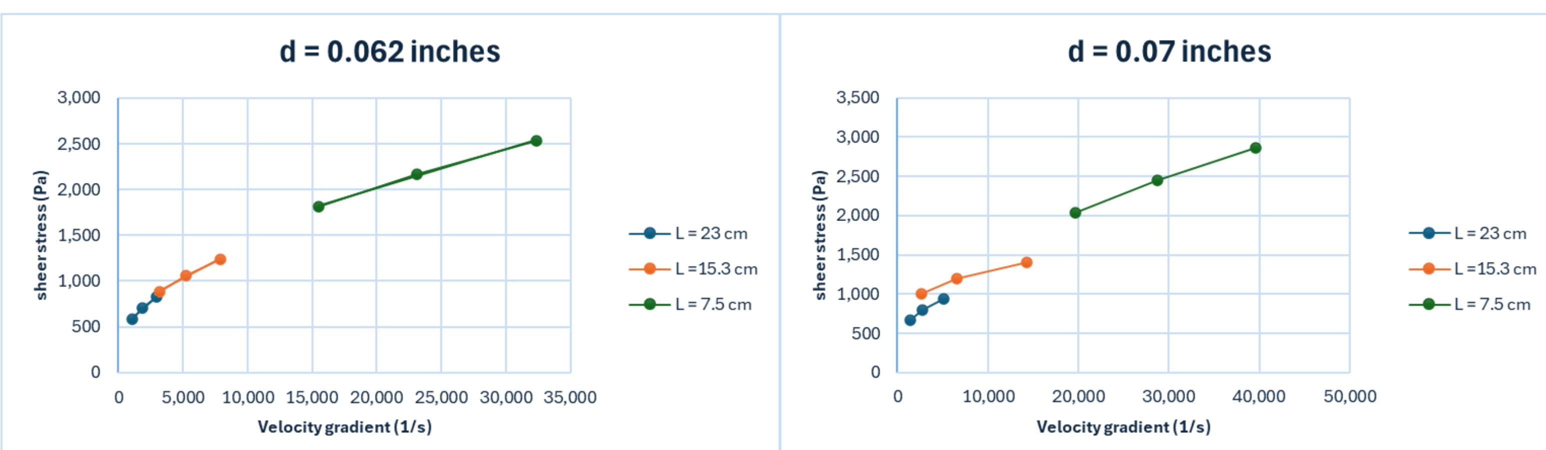
5. Discussion (15 pts)

The main objective of this lab was to calculate the n and m values in our power law analytical equation for the flowrate of the 1 wt% Carbopol solution. Across all six capillary tubes used in the experiment, we calculated an n value equal to ~0.399, as shown in the graph from Figure 1:



These results are very similar to those found in *Rheological Characterization of Carbopol® Dispersions in Water and in Water/Glycerol Solutions* by Varges, Priscilla R. et al [4]. In this study, they used a rheometer to measure multiple different wt% Carbopol solutions ranging from .1wt% to .3wt% and found that the n value fell as low as .384 and as high as .48. Our data point of n=.399 falls into this range, signifying that our results are accurate, at least in comparison to this study.

With an n value <1, we concluded that Carbopol is a shear-thinning material, meaning that the viscosity decreases as more shear is applied to it. This claim is supported by the graphs in Figure 2:



There is a noticeable pseudoplastic behavior happening for each length on the graph. Rather than adjusting linearly, the Shear Stress experienced by the Carbopol solution tapers off as you increase the velocity gradient. This points to the material being viscoelastic/sheer thinning. These results agree with literature as well:

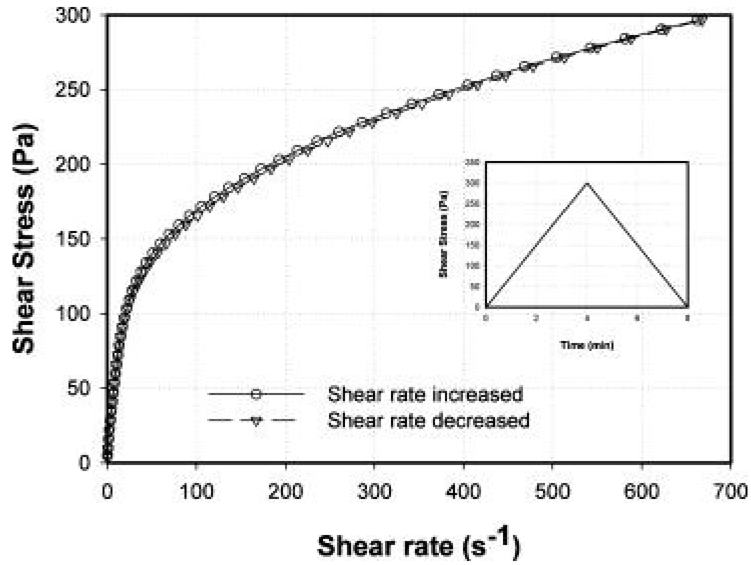


Figure 3: Sheer Stress vs Shear Rate for a .35wt% Carbopol Solution

In *Rheological properties of Carbopol containing nanoparticles* by Baek, Gookhyun et al. [5], researchers did similar rheology testing on Carbopol solutions ranging from .1wt% to 1.5wt%. As shown in Figure 3, when varying the shear rate or Shear Stress vs. Shear Rate (velocity gradient), there is a noticeable tapering relationship between the two, indicating
Shear thinning properties of the material. These results agree with our results.

Something important to mention is the magnitude of the velocity gradient in our testing versus others. Many papers conduct rheology/viscosity measurements with industry-standard rheometers, such as a Parallel Plate Rheometer (AR2000) which was used to achieve the results in Figure 3. With these pieces of equipment, one can explore the effects of very low shear rates and need not go to extremely high rates to draw conclusions. Our experiment utilized a capillary tube viscometer, which while effective and cheaper, can not explore very low shear rates. This is because applying a low pressure drop to the system results in a very small – or no flowrate. Without adequate flow, we can not back calculate our velocity gradient in the time allotted for the lab. Thus, we had to choose large pressure drops that gave us reasonable flowrates, saving us time and enabling good data collection. The consequence of this is very high values for our velocity gradient.

We can definitively conclude that our results prove that a 1wt% Carbopol solution is sheer thinning. This conclusion agrees with our data and with literature, stating that the relationship between Shear Rate and Shear Stress is pseudoplastic. However, we cannot quite conclude that the magnitude of our data is correct. This is because we were limited to very high pressure drops and additionally to unavoidable human & machine error. One area where accuracy errors occurred was flowrate calculations. For each capillary tube, we set 3 different pressure drops and measured how long it took to fill a container with 20 grams of Carbopol solution. Error with timing, shutting off the valve, and with the tare of the scale all contribute to not getting a 100% accurate result. Additionally, there may have been some transient flow happening during measurement that was not accounted for. For each capillary tube set at a given pressure, we would only collect one data point unless we suspected something wrong. It would be better if we ran 2-3 trials for every single data point to ensure there is not transient flow at any point of the data collection. This would account for the precision and accuracy of our results.

Two final points that could contribute to error is the starting solution pH and wt% and the calculation for density. When we originally created our 1wt% Carbopol solution, there could have been error in weighing out the precise amount of powder or water needed to reach the 1wt% threshold. Additionally, we needed to add a lot of sodium hydroxide to make the solution basic (in the pH range of 5-8) for it to begin thickening. This is a large range for the pH and generally we would want to account for the specific value of the pH when presenting the magnitude of our results. Lastly, the density was only calculated once, using non-sheared solution. It could be that shearing the Carbopol solution could change the behavior of the density, such as eliminating air bubbles, adding more air bubbles, or simply changing the packing of the Carbopol. By only measuring the density before the trial, we can not explore this additional nuance which could exist in the system.

6. References (10 pts)

[1] Department of Chemical Engineering, *Viscosity of a Non-Newtonian Fluid Lab Manual*, University of Illinois at Chicago, 2025.
[2] R.B. Bird, W.E. Stewart, and E.N. Lightfoot, *Transport Phenomena*, 2nd ed. John Wiley and Sons, Inc., 2002.
[3] NBCHAO. (n.d.). Working Principle, Classification and Application of Capillary Viscometer. NBCHAO. https://en.nbchao.com/k/14058/

[4] R. Varges, Priscilla, Camila M. Costa, Bruno S. Fonseca, Mônica F. Naccache, and Paulo R. De Souza Mendes. "Rheological Characterization of Carbopol® Dispersions in Water and in Water/Glycerol Solutions." MDPI, January 4, 2019. https://www.mdpi.com/2311-5521/4/1/3.
[5] Rheological properties of Carbopol containing nanoparticles | journal of rheology | AIP publishing. Accessed September 16, 2025.

https://pubs.aip.org/sor/jor/article-abstract/55/2/313/240607/Rheological-properties-of-Carbopol-containing?redirectedFrom=fulltext.

7. Appendix I: Data Tabulation/Graphs (10 pts)

As mentioned in the "Results" section, the length and diameter of each capillary was measured before conducting any experiments, and the pressure drop was set with a regulator. The density of the 1 wt% Carbopol solution was measured once, which was then used to calculate volume by dividing mass by it. The only values measured during each run were the weight of Carbopol and time elapsed. The data collected during the experiment is compiled in the table below.

run #	time (s)	pressure drop (pisg)	Weight (g)	length(cm)	diameter(inches)	Volume (mL)
1	72.06	50	20.23	23	0.062	19.981
2	42.2	60	20.29	23	0.062	20.040
3	26.73	70	20.4	23	0.062	20.149
4	23.2	50	20.33	15.3	0.062	20.080
5	14.21	60	20.57	15.3	0.062	20.317
6	9.6	70	20.69	15.3	0.062	20.435
7	4.43	50	20.78	7.5	0.062	20.524
8	3.18	60	21.62	7.5	0.062	21.354
9	2.1	70	20.55	7.5	0.062	20.297
trial for 10	180	50	20.27	23	0.07	20.020
10	47.77	50	20.47	23	0.07	20.218
trial for 11	21.1	60	20.44	23	0.07	20.188
11	20.31	60	20.51	23	0.07	20.257
12	13.49	70	20.63	23	0.07	20.376
trial for 13	25.62	50	20.5	15.3	0.07	20.247
13	29.34	50	20.52	15.3	0.07	20.267
14	9.77	60	20.94	15.3	0.07	20.682
15	5.7	70	20.31	15.3	0.07	20.060
16	2.38	50	20.3	7.5	0.07	20.050
17	1.48	60	19.38	7.5	0.07	19.141
18	1.16	70	19.78	7.5	0.07	19.536

Raw Data

In order to use this data meaningfully, it had to be converted to SI units. The calculations done with the converted data is described in the "Results" section, and the data and calculations are compiled in the table below. Note that since most values are extremely small and only differ slightly, many significant figures had to be shown.

run #	time (s)	pressure drop (Pa)	Weight (kg)	length(m)	radius(m)	Volume (m^3)	Q (m^3/s)	ln(Q/pi*R^3)	ln(deltaP*R/2L)	L/D ratio	n	m	sheer stress (Pa)	velocity gradient (1/s)
1	72.06	344737.85	0.02023	0.23	0.0007874	0.00001998	2.77E-07	5.20	6.38	146.05	0.3365	56.28	590.10	1078.64
2	42.2	413685.42	0.02029	0.23	0.0007874	0.00002004	4.75E-07	5.74	6.56	146.05	0.3365	56.28	708.12	1854.33
3	26.73	482632.99	0.02040	0.23	0.0007874	0.00002015	7.54E-07	6.20	6.72	146.05	0.3365	56.28	826.14	2931.82
4	23.2	344737.85	0.02033	0.153	0.0007874	0.00002008	8.65E-07	6.34	6.79	97.16	0.3734	43.47	887.08	3219.01
5	14.21	413685.42	0.02057	0.153	0.0007874	0.00002032	1.43E-06	6.84	6.97	97.16	0.3734	43.47	1064.50	5245.38
6	9.6	482632.99	0.02069	0.153	0.0007874	0.00002044	2.13E-06	7.24	7.12	97.16	0.3734	43.47	1241.91	7926.23
7	4.43	344737.85	0.02078	0.075	0.0007874	0.00002052	4.63E-06	8.01	7.50	47.63	0.4577	21.85	1809.64	15519.58
8	3.18	413685.42	0.02162	0.075	0.0007874	0.00002135	6.71E-06	8.38	7.68	47.63	0.4577	21.85	2171.57	23114.16
9	2.1	482632.99	0.02055	0.075	0.0007874	0.00002030	9.67E-06	8.75	7.84	47.63	0.4577	21.85	2533.50	32370.24
10	47.77	344737.85	0.02047	0.23	0.0008890	0.00002022	4.23E-07	5.26	6.50	129.36	0.2567	104.32	666.24	1370.96
11	20.31	413685.42	0.02051	0.23	0.0008890	0.00002026	9.97E-07	6.11	6.68	129.36	0.2567	104.32	799.49	2789.22
12	13.49	482632.99	0.02063	0.23	0.0008890	0.00002038	1.51E-06	6.53	6.84	129.36	0.2567	104.32	932.74	5084.89
13	29.34	344737.85	0.02052	0.153	0.0008890	0.00002027	6.91E-07	5.75	6.91	86.05	0.1994	208.20	1001.54	2637.78
14	9.77	413685.42	0.02094	0.153	0.0008890	0.00002068	2.12E-06	6.87	7.09	86.05	0.1994	208.20	1201.85	6581.66
15	5.7	482632.99	0.02031	0.153	0.0008890	0.00002006	3.52E-06	7.37	7.25	86.05	0.1994	208.20	1402.16	14258.59
16	2.38	344737.85	0.02030	0.075	0.0008890	0.00002005	8.42E-06	8.25	7.62	42.18	0.4799	17.77	2043.15	19669.57
17	1.48	413685.42	0.01938	0.075	0.0008890	0.00001914	1.29E-05	8.68	7.80	42.18	0.4799	17.77	2451.78	28760.09
18	1.16	482632.99	0.01978	0.075	0.0008890	0.00001954	1.68E-05	8.94	7.96	42.18	0.4799	17.77	2860.40	39654.44

Table 2: Data and Calculations

8. Appendix II: Error Analysis (10 pts)

The main source of error in calculating the flow rates comes from measuring the time and mass of each run. The average runs lasted around 70-100 seconds the stopwatch error was about 0.1 seconds or 0.2% error. The balance we used was precise to ±0.01g and the samples were around 20g, so this will give us an error of only about 0.05%. So, in general the errors are very small which means that our timing and mass measurements almost had no impact on the flow rate results. Time measurement error: $\frac{0.1}{70}=0.14\%\ to\ \frac{0.1}{100}=0.1\%$. So, timing error is less than 0.2% uncertainty. Mass measurement error is $\frac{0.01}{20}=0.05\%$ So, the scale is even more precise relative to the sample size. Combined error in flow rate $Q=\frac{V}{t}$, where Q is the volumetric flow rate. Since Q depends on both volume and time, the small uncertainties for each measurement will be combine $Total\ error=\sqrt{(0.2\%)^2+(0.05\%)^2}=0.21\%$.

We set ΔP with the regulator at the start of each run, but during longer experiments the pressure dropped slightly over time, and we did not record this change. This made our ΔP values somewhat uncertain, which mostly affects the consistency index (m). The flow behavior index (n) was more consistent, since it depends less on the absolute pressure readings.

We also assumed the fluid density was constant, but temperature, shear, and pressure can cause small changes which will lead to small systematic errors in calculated volumes and flow rates. We also assumed no wall slip condition, but Carbopol will often slip on smooth tube walls, and if that happened the actual shear stress would be lower than calculated making the fluid appear more shear thinning than it really is.
Other possible sources of error include human reaction time when using the stopwatch, which could add one or two seconds of delay, and we also ignored gravity effects. Since the tubes were short and elevation wasn't that high ignoring gravity was a reasonable assumption.

9. Appendix III: Sample Calculations (10 pts)

Before data analysis, many unit conversions had to be made, since all our significant equations required input in SI units. I used the first run of Table 1 for the following conversions:

- Pressure Drop: $psig \rightarrow Pascal$

$$50\,psig\,x\,6{,}894.757\,\frac{Pa}{psig}=344{,}737.85\,Pa$$

- Mass: $grams \rightarrow kilograms$

$$20.23\,g\,x\,\frac{1\,kg}{1000\,g}=0.02023\,kg$$

- Length: $centimeters \rightarrow meters$

$$23\,cm\,x\,\frac{1\,m}{100\,cm}=0.23\,m$$

- Diameter to Radius: $inches \rightarrow meters$

$$0.062 \in x\,\frac{0.0254\,m}{¿}\,x\,\frac{1}{2}=0.0007874\,m$$

- Volume: $mililiter \rightarrow meter\,cubed$

$$19.981\,mL\,x\,\frac{1\,m^3}{1000000\,mL}=0.000019981\,m^3$$

With all necessary unit conversions completed, let's begin calculating the properties necessary to determine the type of fluid that our solution belongs to. All units for the following calculations are in SI. First, we must calculate the flow rate:

$$Q=\frac{V}{t}=\frac{0.000019981\,m^3}{72.06\,s}=2.77\,x\,10^{-7}\,\frac{m^3}{s}$$

With the flow rate determined, we can calculate the log-log axes for power-law fit. Note that despite the values inside the parenthesis having units, the plotted values that have been logged are dimensionless. The calculations are shown below:

$$x=\log\left(\frac{Q}{\pi R^3}\right)=\log\left(\frac{2.77\,x\,10^{-7}\,\frac{m^3}{s}}{\pi*(0.0007874\,m)^3}\right)=5.20$$

$$y=\log\left(\frac{\Delta PR}{2L}\right)=\log\left(\frac{344{,}737.85\,Pa*0.0007874\,m}{2*0.23\,m}\right)=6.38$$

This set of calculations is repeated twice more for each capillary tube. Next, I will calculate the sheer stress and sheer rate. Each tube has a different *n* and *m* value. The calculations are as follows:

$$\tau_W=\frac{\Delta P}{2L}r=\frac{344{,}737.85\,Pa}{2*0.23\,m}0.0007874\,m=590.10\,Pa$$

$$\frac{d v_z}{dr} = \left(\frac{1}{m}\frac{\Delta P}{2L}\right)^{1/n} r^{1/n} ¿ \left(\frac{1}{56.28}\frac{344{,}737.85\,Pa}{2*0.23\,m}\right)^{1/0.3365} (0.0007874\,m)^{1/0.3365} = 1078.64\,s^{-1}$$

This calculation is repeated for each capillary tube, which is then graphed. This concludes all sample calculations.

10. Appendix IV: Individual Team Contributions

NAME: **Joshua Olusoji**

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Present both days of lab. Helped with all parts of the lab, including making the Carbopol solution, conducting trials, and collecting data
Pre-Lab Editing	0.5	Reviewed and cleaned up this section.
Experimental Objective (Prelab)	0.5	Reviewed and cleaned up this section.
Apparatus (prelab)	2	Completed this section in its entirety.
Final Lab Editing	0.5	Grammer, structure, formatting
Results (final lab)	1.5	Helped clear up errors in the results and post-lab calculations & graphs.
Discussion (final lab)	4	Researched literature on Carbopol solutions and completed the Discussion section in its entirety.
References (final and/or prelab)	0.5	Created my own citations and ensured they were referenced properly.
Data Tabulation/Graphs (final)	1	Helped clear up errors in the graphs formatting.
Total	14.5	

NAME: **Omar Zaruk**

	TIME (HOURS)	DESCRIPTION

Operator (Both Lab Days)	4	Operated and assisted both days
Pre-Lab Editing	0.25	Refined formating
Procedure (prelab)	3	Completed the entirety of the experiment plan
Final Lab Editing	0.5	Formatting, fixing errors
Introduction (Final lab)	2	Completed entirely of this section
Theory (Final Lab)	4	Completed entirely of this section
References (final and/or prelab)	0.1	Added 3rd reference
Power Point Presentation	0.5	Refining
Total	14.35	

NAME: **Sanad Abu Hweij**

	TIME (HOURS)	**DESCRIPTION**
Operator (Both Lab Days)	4	Operator during both lab days ran equipment, collected samples, timed runs
Pre-Lab Editing	1	Pre-lab editing
Experimental Objective (Prelab)	0.5	Wrote experimental objective
Apparatus (prelab)	1	Described apparatus setup viscometer, pressure gauge, tubing
Project Safety Evaluation (prelab)	0.5	Listed materials and supplies for prelab
Executive Summary (final lab)	1.5	Wrote executive summary for final lab
Theory (Final Lab)	0.5	Reviewed theory and equations
Results (final lab)	0.5	Checked data tables and figures after they were made
Data Tabulation/Graphs (final)	0.5	Looked over tables to check accuracy
Error Analysis (final lab)	2	Wrote error analysis timing, mass, pressure, density, wall slip
Sample Calculations (final	0.5	Checked sample calculations

Lab)		for accuracy
Power Point Presentation	0.5	Refining
Total	13	Total hours

NAME: **Qussay Suleiman**

	TIME (HOURS)	**DESCRIPTION**
Operator (Both Lab Days)	2	Performed and assisted with operations both days
Pre-Lab Editing	0.5	Briefly edited the prelab before submission
Project Safety Evaluation (prelab)	2	Completed the entirety of this section
Final Lab Editing	1	Checked for typos and other minor structuring errors
Results (final lab)	4	Completed the entirety of this section
References (final and/or prelab)	0.1	Added the first 2 references
Data Tabulation/Graphs (final)	1	Completed the entirety of this section
Sample Calculations (final Lab)	2	Completed the entirety of this section
Power Point Presentation	3	Made the first draft of the slideshow
Total	16.6	Completed a sizable amount of work for this lab